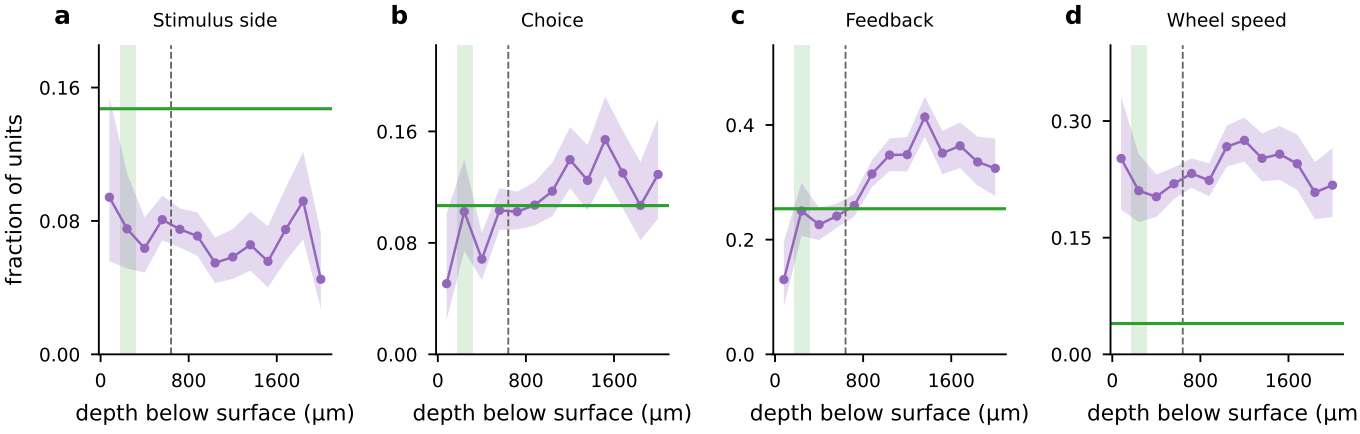

Figure 1: Depth sensitivity of the BWM comparison. Green line: mesoscope fraction over all neurons; green shading: mesoscope imaging depth (174–318 μm, 5–95 % of the histology-registered sessions); dashed line: the 640 μm cutoff of the depth-limited BWM set. Depth: Allen CCF top surface at the unit's ML-AP position minus its DV. **a** Fraction of BWM isocortex units tuned to stimulus side ($p < 0.05$, same test as the maps) per 160 μm depth band (purple; band: 95 % Wilson interval; bands with $\geq$30 units): 7.5% of BWM units within 640 μm, 6.7% deeper, mesoscope 14.7%. **b** Choice: 9.0% of BWM units within 640 μm, 11.4% deeper, mesoscope 10.7%. **c** Feedback: 23.3% of BWM units within 640 μm, 31.2% deeper, mesoscope 25.4%. **d** Fraction of units correlated with wheel speed ($r > 0.1$ over the wheel bins of both the even and the odd trials): 21.5% of BWM units within 640 μm, 23.2% deeper, mesoscope 4.0%.